MIL-T-52176B
26 February 1976
SUPERSEDING
MIL-T-52176A
4 November 1969

MILITARY SPECIFICATION

TEETH, SURFACE RIPPING; (POINTS, SHANKS, KEEPERS AND KEEPER LOCKS), BUCKET AND DIPPER, CRANE-SHOVEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers points, shanks, keepers, and keeper locks which make up power excavator bucket and dipper teeth.

1.2 Classification.

1.2.1 Points. The points shall be of the following sizes, as specified (see 6.2):

Size 3 - For use on 3/8- and 1/2-cubic-yard buckets.
Size 4 - For use on 3/4-, 1-, and 1-1/2-cubic-yard buckets.
Size 5 - For use on 2- and 2-1/2-cubic-yard buckets.
Size 6 - For use on 3-1/2-cubic-yard buckets.

1.2.2 Shanks. The shanks shall be of the following types and sizes, as specified (see 6.2):

Type I - For clamshell buckets.

Size 3 - For 3/8- and 1/2-cubic-yard buckets.
Size 4 - For 3/4-, 1-, and 1-1/2-cubic-yard buckets.
Size 5 - For 2- and 2-1/2-cubic-yard buckets.
Size 6 - For 3-1/2-cubic-yard buckets.

Type II - For backhoe, dipper, and dragline buckets.

Size 3 - For 3/8- and 1/2-cubic-yard buckets.
Size 4 - For 3/4-, 1-, and 1-1/2-cubic-yard buckets.
Size 5 - For 2- and 2-1/2-cubic-yard buckets.
Size 6 - For 3-1/2-cubic-yard buckets.

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1.2.3 Keepers. The keepers shall be of the following sizes, as specified (see 6.2):

Size 3 - For 3/8- and 1/2-cubic-yard buckets.
Size 4 - For 3/4-, 1-, and 1-1/2-cubic-yard buckets.
Size 5 - For 2- and 2-1/2-cubic-yard buckets.
Size 6 - For 3-1/2-cubic-yard buckets.

1.2.4 Keeper locks. The keeper locks shall be of the following sizes, as specified (see 6.2):

Size 3 - For 3/8- and 1/2-cubic-yard buckets.
Size 4 - For 3/4-, 1-, and 1-1/2-cubic-yard buckets.
Size 5 - For 2- and 2-1/2-cubic-yard buckets.
Size 6 - For 3-1/2-cubic-yard buckets.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

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DRAWINGS

ME

TA13213E6846	- 3 Inch Point and Shank Assembly Type I, For Clamshell Bucket.
TA13213E6847	- 4 Inch Point and Shank Assembly Type I, For Clamshell Bucket.
TA13213E6848	- 5 Inch Point and Shank Assembly Type I, For Clamshell Bucket.
TA13213E6849	- 6 Inch Point and Shank Assembly Type I, For Clamshell Bucket.
TA13213E6850	- 3 Inch Point and Shank Assembly Type II, For Backhoe, Dipper and Dragline Bucket.
TA13213E6851	- 4 Inch Point and Shank Assembly Type II, For Backhoe, Dipper and Dragline Bucket.
TA13213E6852	- 5 Inch Point and Shank Assembly Type II, For Backhoe, Dipper and Dragline Bucket.
TA13213E6853	- 6 Inch Point and Shank Assembly Type II, For Backhoe, Dipper and Dragline Bucket.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The complete tooth shall consist of one each of the following: Points, shank, keeper, and keeper lock, fully assembled. Type I teeth shall be as shown on Top Assemblies TA13213E6846, TA13213E6847, TA13213E6848, and TA13213E6849, as applicable. Type II teeth shall be as shown on Top Assemblies TA13213E6850, TA13213E6851, TA13213E6852, and TA13213E6853, as applicable. Tooth components may be procured separately and shall be as shown on the respective drawings for the size and type of tooth required.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the tooth or tooth components. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first-produced tooth or tooth components). The contractor shall furnish a complete tooth or tooth components, as applicable (see 3.1), for examination and test within the time frame specified (see 6.2) to prove that his production methods will produce teeth or tooth components, as applicable, that comply with the requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as shown on the applicable drawings.

3.4 Identification marking. The equipment shall be identified in accordance with MIL-STD-130.

3.5 Interchangeability. All like components shall be functionally and dimensionally interchangeable.

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3.6 Workmanship. All parts, components, and assemblies including castings, forgings, molded parts, stampings, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required. Castings shall be sound and free from patching, misplaced coring, warping, or any other defects which might render the casting unsound for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced complete tooth or tooth components inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.5).

4.3 First-produced complete tooth or tooth components inspection.

4.3.1 Examination. The first-produced complete tooth or tooth components, as applicable, shall be examined as specified in 4.4.2, except that Item 106 shall not apply to this examination. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling of tooth components for examination shall be in accordance with MIL-STD-105.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the following major defects. AQL shall be 4.0 percent defective.

101. Dimensions not as specified.
102. Material not as specified.
103. Any mechanical property not as specified.
104. Identification markings not as specified.
105. Evidence of cracks, rust, or other defects that could impair serviceability.
106. Tooth components not interchangeable with other like components or not interchangeable with like components of the first-produced complete tooth or tooth components.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107. Materials and containers not as specified for Level A or B. Each incorrect material or container shall constitute one defeat.
108. Unprotected surfaces not coated with preservative as specified.
109. Items of unlike description packed together for Level A or B.
110. Gross weight exceeds the limitations of the box specification for Level A or B.
111. Blocking and bracing not in accordance with the referenced document for Level A or B.
112. Strapping not as specified for Level A.
113. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation (see 6.4). Unprotected surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

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5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Tooth assemblies. The tooth assemblies of like description shall be packed together in close-fitting boxes conforming to PPP-B-621, Class 2, style optional, or in boxes conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the box. The contents shall be blocked and braced within the box to prevent movement in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. The strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable; and, unless otherwise specified by the ordering activity, shall be Finish B. When specified (see 6.2), Finish A strapping shall be used.

5.2.1.2 Points, shanks, keepers, and keeper locks. Points, shanks, keepers, and keeper locks of like description shall be packed together as specified in 5.2.1.1 for tooth assemblies.

5.2.2 Level B. Level B packing shall be as specified in 5.2.1 for Level A, except that boxes shall be Domestic Type or Class, as applicable. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. The tooth assemblies or the component parts shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings, in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Shipping containers shall be marked in accordance with MIL-STD-129 (see 6.4).

6. NOTES

6.1 Intended use. The points, shanks, keepers, and keeper locks form component parts of bucket teeth which are intended for use on all buckets (3/8- through 3-1/2-cubic-yard capacity) for clamshell, dragline, backhoe, and shovel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of points required (see 1.2.1).
(c) Type and size of shanks required (see 1.2.2).
(d) Size of keepers required (see 1.2.3).
(e) Size of keeper locks required (see 1.2.4).

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(f) Time frame required for submission of first-produced tooth or tooth components, as applicable (see 3.2).
(g) Level of packing required (see 5.2).
(h) When Finish A strapping is required (see 5.2.1.1).

6.3 First-produced tooth or tooth components. Any changes or deviations of production teeth or tooth components from the approved first-produced tooth or tooth components during production will be subject to the approval of the contracting officer. Approval of the first-produced tooth or tooth components will not relieve the contractor of his obligation to furnish tooth or tooth components conforming to this specification.

6.4 Levels of preservation have not been included as the preservation specified is the minimum acceptable for any level. For purposes of preservation/packing level marking, the boxes shall be marked A/A, A/B, or A/C, as applicable.

Custodians:

Army - ME
Navy - YD

User activities:

Army - CE
Navy - MC

Preparing activity:

Army - ME

Project No. 3815-0081

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DOCUMENT IDENTIFIER AND TITLE MIL-T-52176B - TEETH, SURFACE RIPPING; (POINTS, SHANKS, KEEPERS AND KEEPER LOCKS), BUCKET AND DIPPER, CRANE-SHOVEL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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